

A5



Tag 3

24.06.2026

Von KI erstellt, von Menschen kontrolliert.

Guten Morgen!

Wie findet Ihr das Logo? ↗

Azure in 5 Wochen

Start 22. Juni 2026 

Introduction to Cloud Computing (AZ-900)

→ AZ-104

- Manage identities and governance in Azure
- Prerequisites for Azure administrators – CloudShell and templates
- Implement and manage storage in Azure
- Configure and manage virtual networks for Azure administrators
- Deploy and manage Azure compute resources
- Monitor and back up Azure resources

Prüfungsvorbereitung, Prüfung

Ende 24. Juli 2026 

Theoria cum Praxi
Wilhelm &
Axelander &
ich

→ Skillable (AZ-900 (Guided Project) AZ-104)

Learning Path LP

API Application
programming Interface

Microsoft Learn profile

AZ-104

Learning Paths:

Microsoft
Graph API

Tenant

EntraID

User

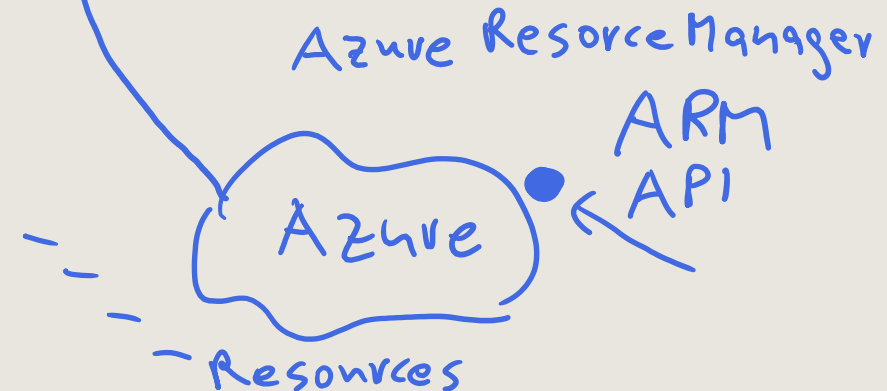
Groups

Devices

App Service
Principal
(SP)

New:
Agent Blueprint?

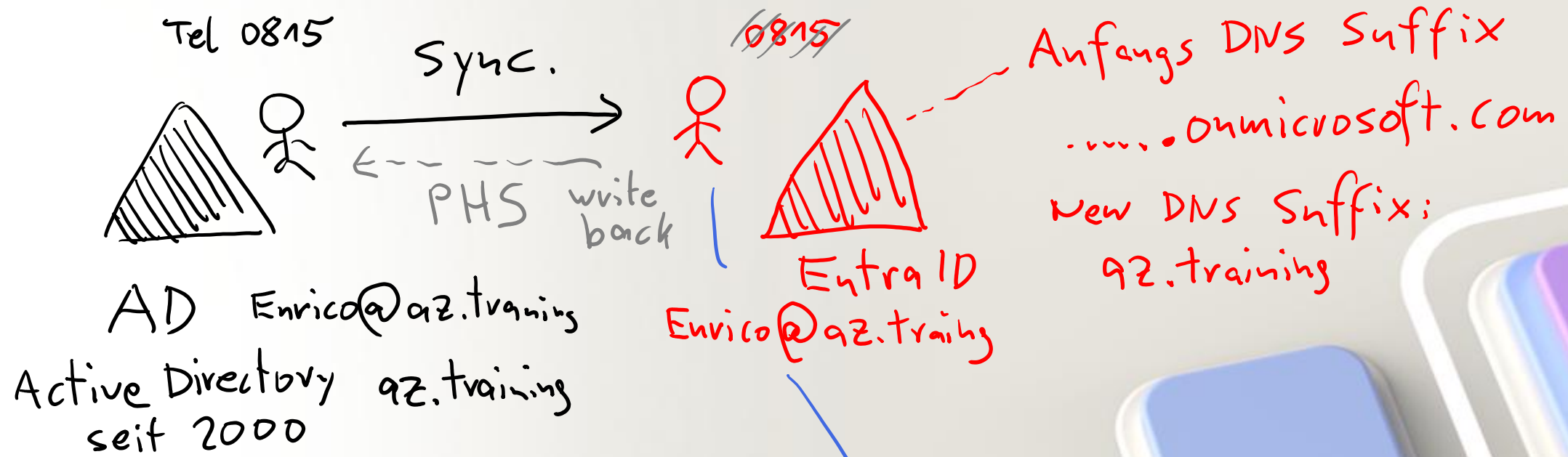
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AZ-104

➔ Manage identities and governance in Azure





Understand Microsoft Entra ID

Hybrid ID  → 

RBAC

Azure

PHS Password Hash Sync.

Wann Sync? alle 30 min.

UPN User Principal Name

Examine Microsoft Entra ID

OAuth 2.0
Open Authorization

Kerberos



- Configure access to applications, including single sign-on
- Manage and provision users and groups
- Providing an identity management solution, including federation
- Implement security features like multi-factor authentication and conditional access

Describe Microsoft Entra ID Concepts

- Entra ID
- AD
- Google
- LinkedIn
- Facebook
- Private MS
- Okta
- GitHub
- [Id Provider]

Concept	Description
Identity ?	An object that can be authenticated
Account	An identity that has data associated with it
Microsoft Entra ID account	An identity created through Microsoft Entra ID or another Microsoft cloud service
Tenant/directory	A dedicated and trusted instance. A tenant is automatically created when your organization signs up for a Microsoft cloud service subscription. • Additional instances can be created • Microsoft Entra ID is the underlying product providing the identity service • The term <i>Tenant</i> means a single instance representing a single organization • The terms <i>Tenant</i> and <i>Directory</i> are often used interchangeably
Azure subscription	Used to pay for Azure cloud services

MFA Multi Factor Auth

Compare Microsoft Entra ID to Active Directory Domain Services



Microsoft Entra ID is primarily an identity solution



Queried using the REST API over HTTP and HTTPS

Microsoft Graph



Uses HTTP and HTTPS protocols such as SAML, WS-Federation, and OpenID Connect for authentication (and OAuth for authorization)



Includes federation services, and many third-party services (such as Facebook)



Microsoft Entra ID users and groups are created in a flat structure, and there are no Organizational Units (OUs) or Group Policy Objects (GPOs)

Compare Microsoft Entra ID plans SKU

Feature	Free	Microsoft ID P1	Microsoft ID P2	Microsoft Entra Suite
Single Sign-On (unlimited)	✓	✓	✓	
Cloud and Federated authentication	✓	✓	✓	
Advanced group management		✓	✓	
Self-service account management portal	✓	✓	✓	
Multifactor authentication (MFA)	✓	✓	✓	
Conditional access CA		✓	✓	
Risk-based Conditional Access (sign-in risk, user risk)			✓	✓
Automated user and group provisioning to apps		✓	✓	✓
Privileged identity management (PIM)			✓	✓
Advanced identity governance				✓

What is self-service password reset in Microsoft Entra ID?

1. Determine who can use self-service password reset
2. Choose the number of authentication methods required and the methods available (email, phone, questions)
3. You can require users to register for SSPR (same process as MFA)

Password reset - Authentication methods
mitanic (Default Directory)

« Save Discard

Diagnose and solve problems

Manage

- 1 Properties
- 2 Authentication methods
- 3 Registration
- Notifications
- Customization
- On-premises integration

Activity

- Audit logs
- Usage & insights

Troubleshooting + Support

- New support request

Number of methods required to reset ⓘ

1 2

Methods available to users

- ☐ Mobile app notification
- ☐ Mobile app code
- ☒ Email
- ☒ Mobile phone
- ☐ Office phone
- ☒ Security questions

Number of questions required to register ⓘ

3 4 5

Number of questions required to reset ⓘ

3 4 5

Select security questions

5 security questions selected

Create, configure, and manage identities



Create User Accounts

 **Users** | All users
Microsoft

[+ New user](#) [+ New guest user](#) [Bulk operations](#) [Refresh](#) [Reset password](#) [Multi-Factor Authentication](#) [Delete user](#)

	Name	User principal name	↑↓	User type	Directory synced
 All users	 Retail Crisis Notifications	████████@microsoft.com		Member	Yes
 Deleted users	 Rumon Sinha	████████@microsoft.onmicrosoft.com		Guest	No
 Password reset	 Momir Radojkovic	████████@microsoft.onmicrosoft.com		Guest	No
 User settings	 Mika Robertson	████████@microsoft.onmicrosoft.com		Member	No
 Diagnose and solve problems					

- All users must have an account ✓

- The account is used for authentication and authorization

- Each user account has additional properties

1. → 2.
Access Token

Get-AzAccessToken

Manage User Accounts

Azure Portal ←
Entra Portal

+ New user + New guest user ↑ Bulk create ↑ Bulk invite ↑ Bulk delete ↓ Download users ↻ Refresh 🔑 Reset password ↗ Multi-Factor Authentication ...

New user

Microsoft



Create user

Create a new user in your organization. This user will have a user name like `alice@Microsoft.onmicrosoft.com`.

[I want to create users in bulk](#)



Invite user

Invite a new guest user to collaborate with your organization. The user will be emailed an invitation they can accept in order to begin collaborating.

[I want to invite guest users in bulk](#)

- Must be Global Administrator or User Administrator to manage users
- User profile (picture, job, contact info) is optional
- Deleted users can be restored for 30 days
- Sign in and audit log information is available

Create Group Accounts

△
Entra ID

△ AD

 Search groups

	Name	Group Type	Membership Type
☐	 Managers	Security	Assigned
☐	 Virtual Machine Administrators	Security	Assigned
☐	 Virtual Network Administrators	Security	Assigned

Group Types

- Security groups
- Microsoft 365 groups

Post fact
Calendar

Membership Types

- Assigned
- Dynamic User
- Dynamic Device (Security groups only)

P2 License needed!

Assign Licenses to Users and Groups

Azure is a cloud service that provides many built-in services for free.

- Microsoft Entra ID comes as a free service
- Gain additional functionality with a P1 or P2 license

Additional Services (like O365 are paid cloud services)

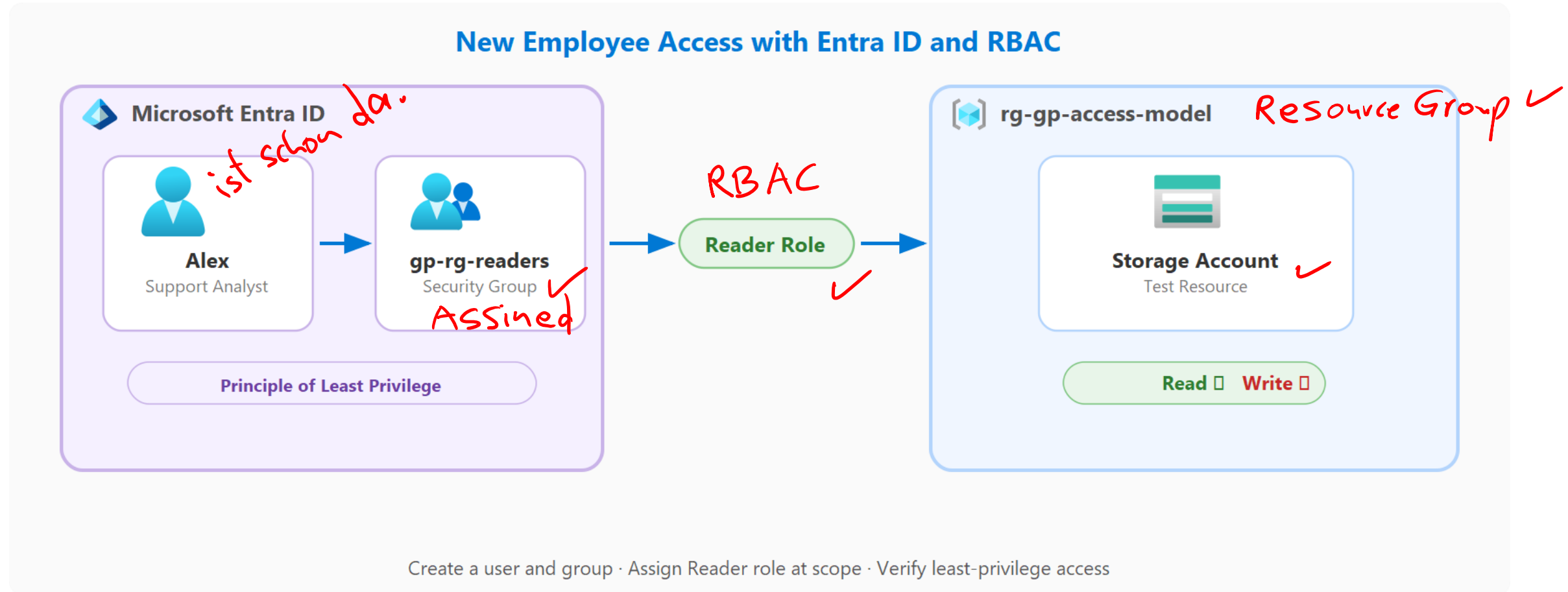
- Microsoft paid cloud services require licenses
- Licenses are assigned to those who need access to the services
- Each user or group requires a separate paid license
- Administrators use the Microsoft 365 Admin portal and Microsoft Graph PowerShell cmdlets to manage licenses

- ☐ View license plans and plan details
- ☐ Set the Usage Location parameter
- ☐ Assign licenses to users and groups
- ☐ Change license plans for users and groups
- ☐ Remove a license

Labs

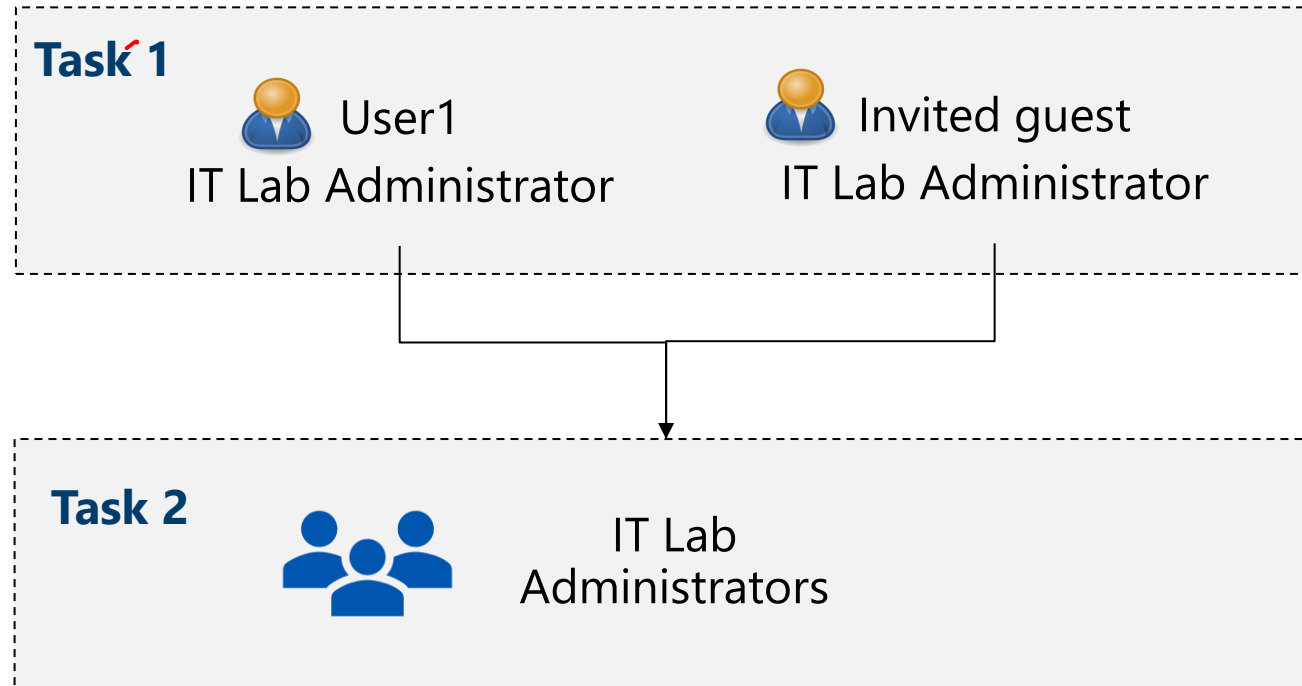


Project: Identity and RBAC onboarding diagram



AZ-104

Lab 01 – Manage Entra ID Identities (architecture diagram)



Tools

API Call

HTTP

Verb GET
Verb POST
PUT
DELETE

CRUD

Azure Portal

New Group

PowerShell
Module
Entra

New-EntraGroup

Graph Explorer

API Call

mit
Access Token

API

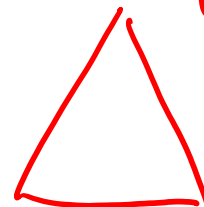
Microsoft Graph

User

Entra ID



Group



Postfach
von Paul
emails

M365

Tenant

ChatGPT

GPT 5.5

Agents

AI App



Bicep
template



Azure Policy Initiatives



Implement Azure Policies

- A service to create, assign, and manage policies
- Runs evaluations and scans for non-compliant resources

Advantages:

- Enforcement and compliance
- Apply policies at scale
- Remediation

Usage Cases

Allowed resource types – Specify the resource types that your organization can deploy

Allowed virtual machine SKUs – Specify a set of virtual machine SKUs that your organization can deploy

Allowed locations – Restrict the locations your organization can specify when deploying resources

Require tag and its value – Enforces a required tag and its value

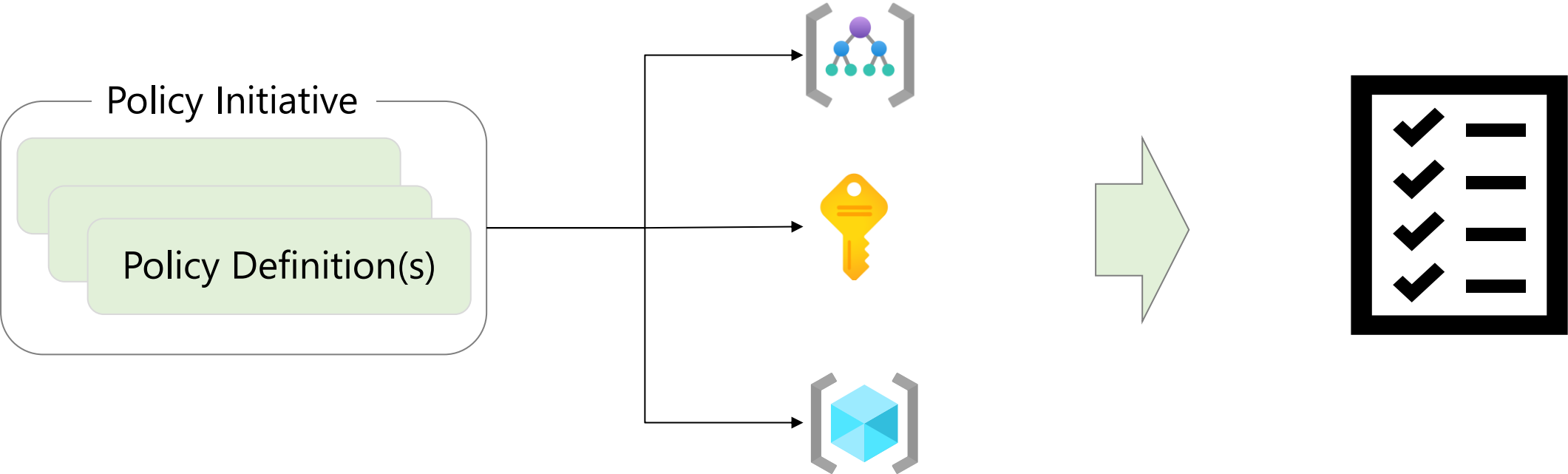
Azure Backup should be enabled for Virtual Machines – Audit if Azure Backup service is enabled for all Virtual machines

Create Azure Policies

Define and create

Scope and assign

Assess compliance



Secure your Azure resources
with Azure role-based access
control (Azure RBAC)



Compare Azure RBAC Roles to Entra ID Roles

RBAC roles provide fine-grained access management

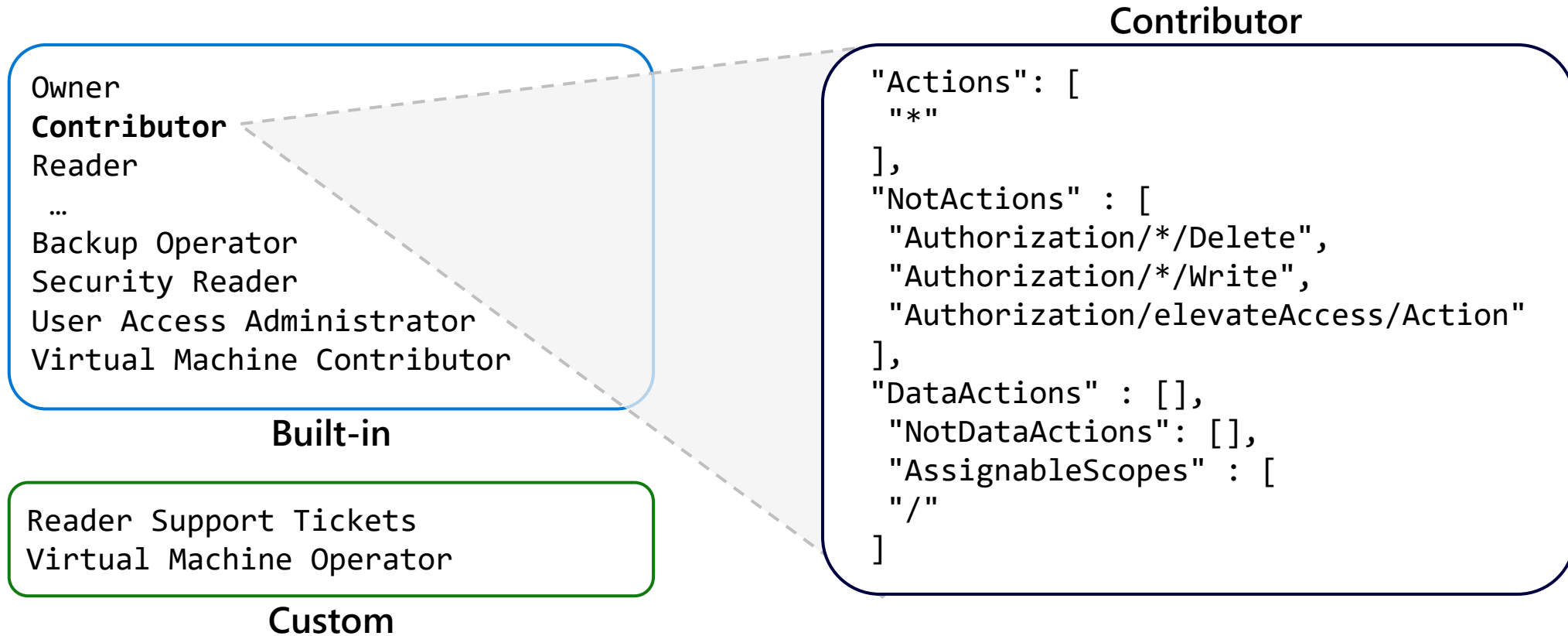
Azure RBAC roles	Entra ID roles
Manage access to Azure resources	Manage access to Entra ID objects
Scope can be specified at multiple levels	Scope is at the tenant level
Role information can be accessed in the Azure portal, Azure CLI, Azure PowerShell, Azure Resource Manager templates, REST API	Role information can be accessed in Azure portal, Microsoft 365 admin portal, Microsoft Graph PowerShell



There are many built-in roles, or you can create your own custom role

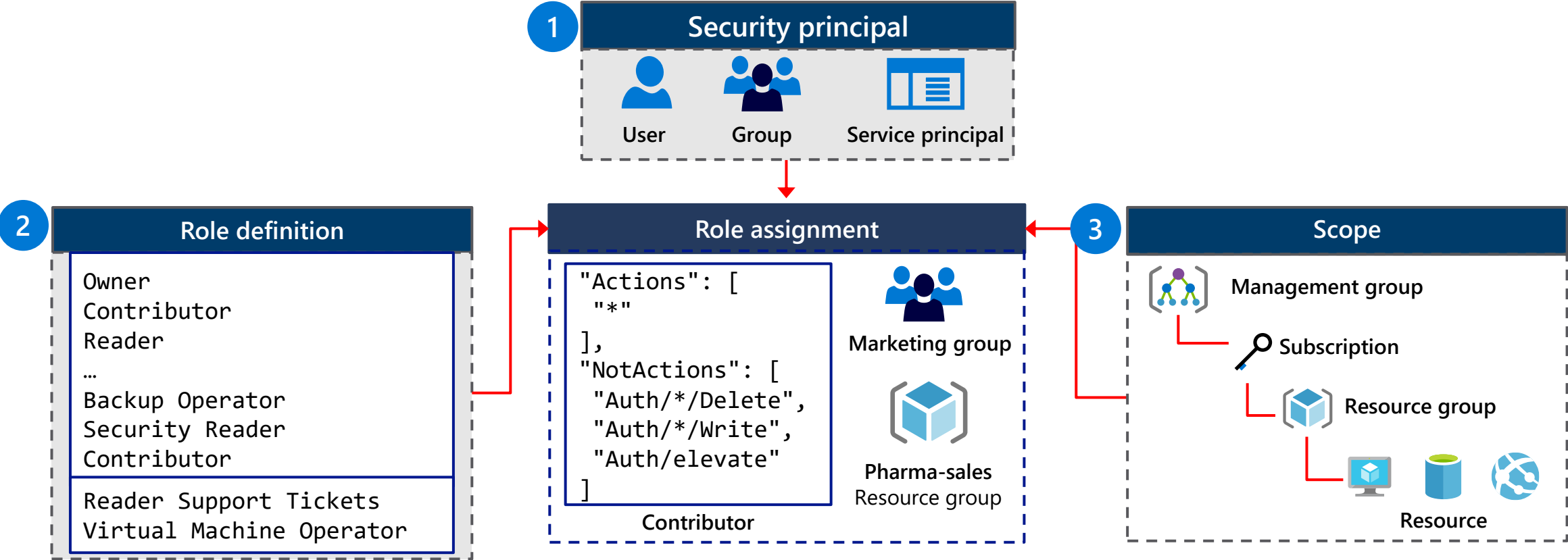
Create a Role Definition

Collection of permissions that lists the operations that can be performed

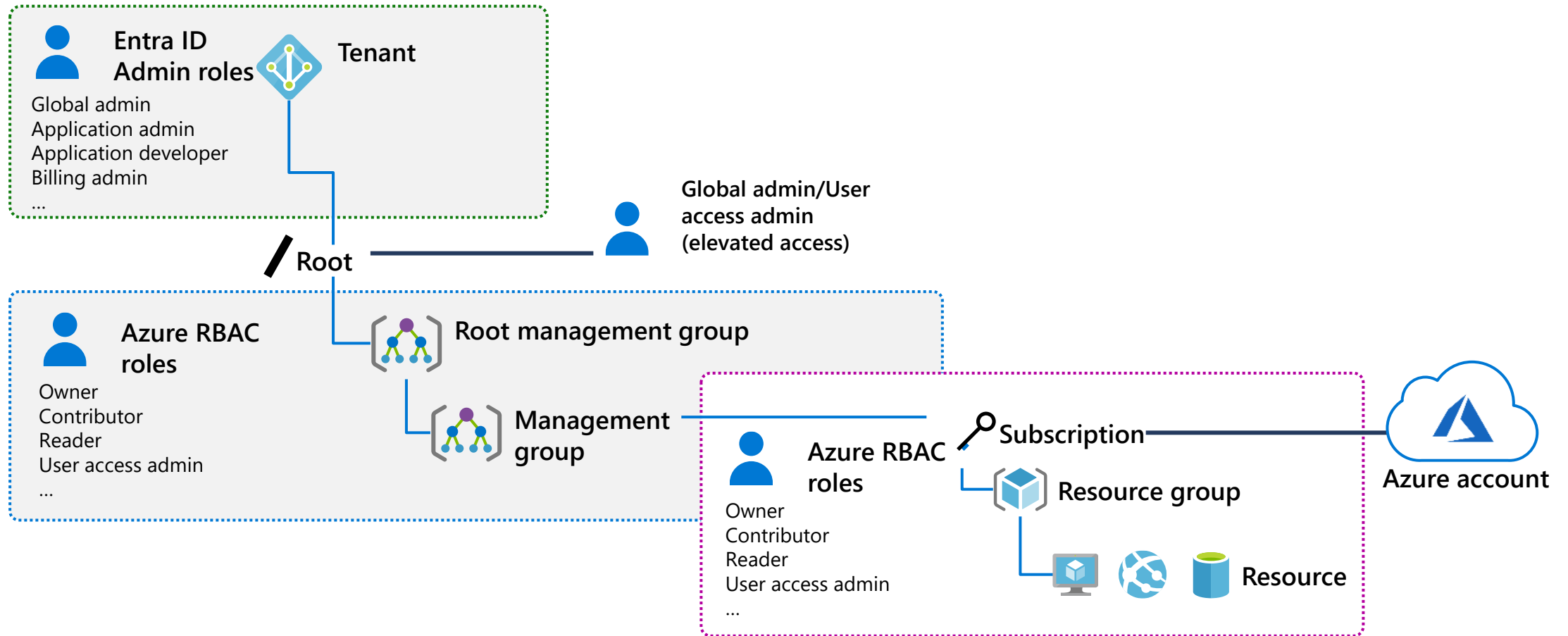


Create a Role Assignment

Process of binding a role definition to a user, group, or service principal at a scope for the purpose of granting access



Apply RBAC Authentication



Labs



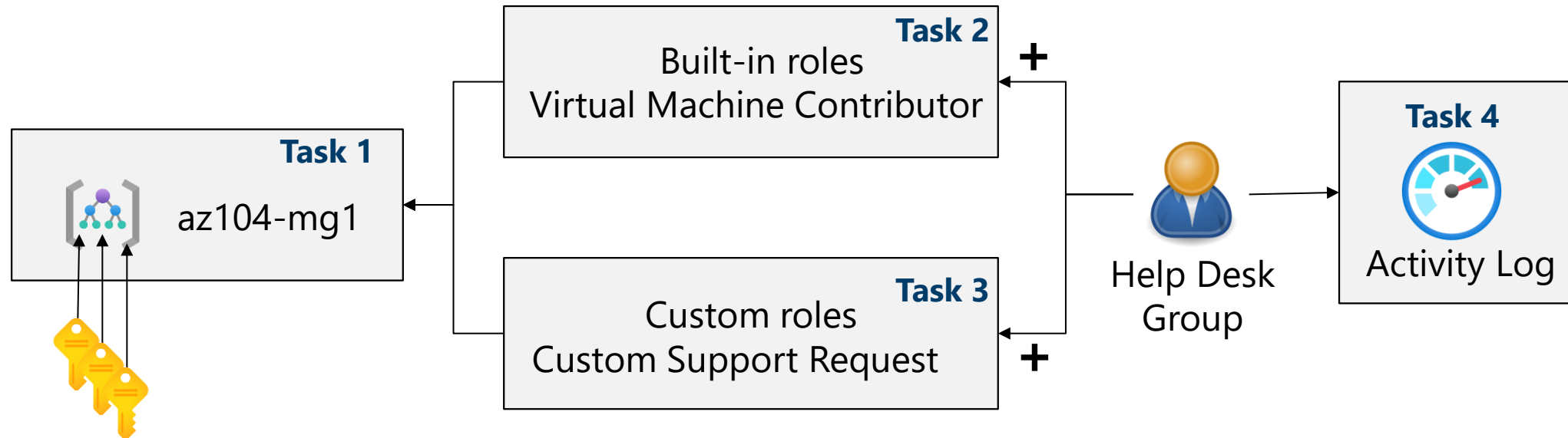
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Lab 02a - Manage Subscriptions and RBAC

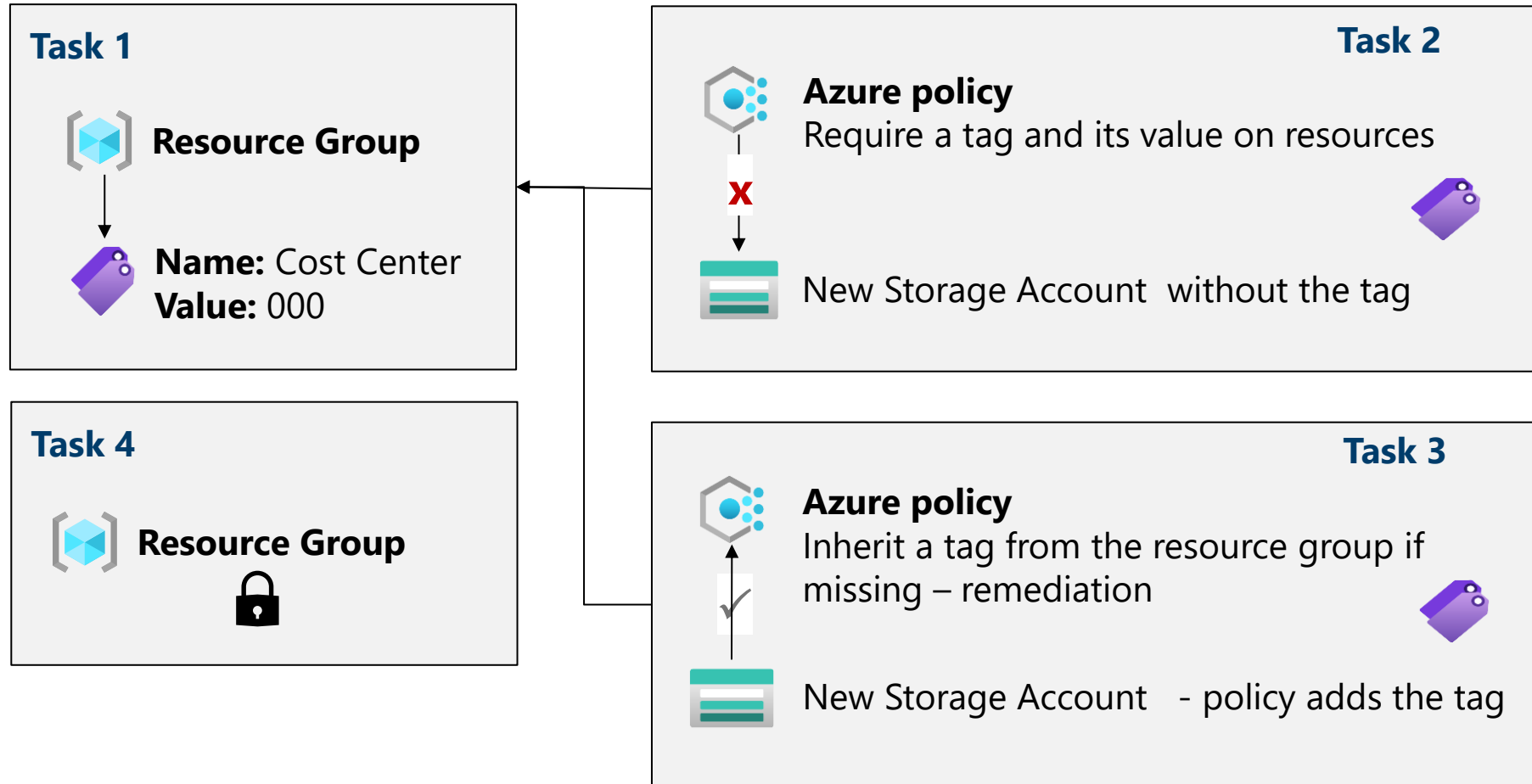
Lab 02b - Manage Governance via Azure Policy



Lab 02a – Architecture diagram



Lab 02b – Architecture diagram



End of presentation



